

Math 527, S10
Exam 2

Name: *Answer*
SSN: xxx-xx-

Instructions:

- There are totally 4 problems.
- You must show all of your work to receive a credit for a correct answer.
- No graphical calculator is allowed in the quiz and exam.

Problem	Points	Score
1	25	
2	25	
3	30	
4	20	
Total	100	

Good Luck !

Problem 1. (25 points) Consider computing the integral $\int_0^1 \cos(2x) dx$.

(a) What is the result of using the composite trapezoidal rule with $h = 1/2$?

Solution: $h = \frac{1}{2}$ gives us a partition of $[0, 1]$
into three points $0, \frac{1}{2}, 1$.

Thus,

$$\begin{aligned}\int_0^1 \cos(2x) dx &\approx \frac{h}{2} [\cos(2 \cdot 0) + 2 \cdot \cos(2 \cdot \frac{1}{2}) + \cos(2 \cdot 1)] \\ &= \frac{1}{4} [1 + 2 \cdot \cos 1 + \cos 2] \\ &\approx 0.4161\end{aligned}$$

(b) If the composite trapezoidal rule is used with $h = 0.01$, what numerical value will the error not exceed?

Solution: We know the error term is given by

$$-\frac{b-a}{12} h^2 f''(\xi) \quad \text{where } \xi \in [0, 1]$$

with $a=0, b=1, h=0.01, f(x) = \cos(2x)$.

$$f'(x) = -2\sin(2x), \quad f''(x) = -4\cos(2x)$$

$$\Rightarrow |f''(\xi)| = |-4\cos(2x)| \leq 4$$

Thus

$$|-\frac{b-a}{12} h^2 f''(\xi)| \leq \frac{1}{12} \cdot (0.01)^2 \cdot 4 = \frac{1}{3} \times 10^{-4}$$

Problem 2. (25 points) Given the following data

x	-1	0	$\frac{1}{2}$	1
y	2	1	0	1

(a) Find a linear spline (piecewise linear) interpolation $S(x)$ for these data.

Solution:

$$S(x) = \begin{cases} -x+1, & -1 \leq x \leq 0 \\ -2x+1, & 0 \leq x \leq \frac{1}{2} \\ 2x-1 & \frac{1}{2} \leq x \leq 1 \end{cases}$$

(b) Verify that

$$Q(x) = \begin{cases} -(x+1)^2 + 2 & -1 \leq x \leq 0 \\ -2x+1 & 0 \leq x \leq \frac{1}{2} \\ 8(x-\frac{1}{2})^2 - 2(x-\frac{1}{2}) & \frac{1}{2} \leq x \leq 1 \end{cases}$$

is a quadratic spline interpolating these data.

Solution: $Q(-1)=2$, $Q(0)=1$, $Q(\frac{1}{2})=0$, $Q(1)=1$

$$\lim_{x \rightarrow 0^-} Q'(x) = \lim_{x \rightarrow 0^-} -2(x+1) = -2 \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{equal}$$

$$\lim_{x \rightarrow 0^+} Q'(x) = \lim_{x \rightarrow 0^+} -2 = -2$$

$$\lim_{x \rightarrow \frac{1}{2}^-} Q'(x) = \lim_{x \rightarrow \frac{1}{2}^-} -2 = -2$$

$$\lim_{x \rightarrow \frac{1}{2}^+} Q'(x) = \lim_{x \rightarrow \frac{1}{2}^+} 16(x-\frac{1}{2}) - 2 = -2 \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{equal}$$

$\Rightarrow Q(x)$ is a quadratic spline interpolating these data.

Problem 3. (30 points) Consider the ordinary differential equation

$$\begin{cases} \frac{dx}{dt} = tx^2 \\ x(1) = 2 \end{cases}$$

(a) Write down the formula of the Euler method for numerically solving this ODE, and then use it to compute $x(1.1)$ with $h = 0.1$.

Solution. Euler's Method:

$$X(t+h) = X(t) + h f(t, X(t)) \quad \text{where } f(t, x) = tx^2.$$

$$\begin{aligned} X(1.1) &= X(1) + 0.1 \cdot f(1, X(1)) \\ &= 2 + 0.1 \cdot 1 \cdot 2^2 \\ &= 2 + 0.4 \\ &= 2.4. \end{aligned}$$

(b) Write down the formula of the Runge-Kutta method of order 2 for numerically solving this ODE, and then use it to recompute $x(1.1)$ with $h = 0.1$.

Solution: RK Method of order 2:

$$X(t+h) = X(t) + \frac{1}{2} k_1 + \frac{1}{2} k_2$$

$$\begin{cases} k_1 = h f(t, X(t)) \\ k_2 = h f(t+h, X(t) + k_1) \end{cases} \quad \text{with } f(t, x) = tx^2$$

$$k_1 = 0.1 \cdot f(1, X(1)) = 0.1 \cdot 1 \cdot 2^2 = 0.4.$$

$$\begin{aligned} k_2 &= 0.1 \cdot f(1.1, X(1) + k_1) \\ &= 0.1 \cdot f(1.1, 2.4) = 0.6336 \end{aligned}$$

$$\begin{aligned} \Rightarrow X(1.1) &= X(1) + \frac{1}{2} k_1 + \frac{1}{2} k_2 \\ &= 2 + \frac{1}{2} \cdot 0.4 + \frac{1}{2} \cdot 0.6336 \\ &= 2.5168 \end{aligned}$$

Problem 4. (20 points) What straight line best fits the following data

x	1	2	3	5
y	0	2	2	4

in the least-squares sense?

Solution: $y = ax + b.$

$$p = \sum_{k=0}^3 x_k = 1 + 2 + 3 + 5 = 11$$

$$q = \sum_{k=0}^3 y_k = 0 + 2 + 2 + 4 = 8$$

$$r = \sum_{k=0}^3 x_k y_k = 1 \cdot 0 + 2 \cdot 2 + 3 \cdot 2 + 5 \cdot 4 = 30$$

$$s = \sum_{k=0}^3 x_k^2 = 1^2 + 2^2 + 3^2 + 5^2 = 39$$

$$\Rightarrow \begin{bmatrix} s & p \\ p & m+1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} r \\ q \end{bmatrix}$$

$$\begin{bmatrix} 39 & 11 \\ 11 & 4 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 30 \\ 8 \end{bmatrix}$$

$$\Rightarrow \begin{cases} a \approx 0.9143 \\ b \approx -0.5142 \end{cases}$$

$$\Rightarrow y = 0.9143x - 0.5142.$$